GLUEABILITY DESCENDS THROUGH ETALE COVERINGS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2487-2498 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2500-2537, the complete author proof (36 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 18 of queue_spaces-pushouts.json (pb=1596, stmt 2487-2498, proof 2500-2537).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0218: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2537–2537. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(R \to R', f)$ be a glueing pair, see above. Let Y be an algebraic space over X. The following are equivalent*

(1) *there exists an étale covering $\{Y_i \to Y\}_{i \in I}$ with Y_i affine and $\Gamma(Y_i, \mathcal{O}_{Y_i})$ glueable as an R-module,*
(2) *for every étale morphism $W \to Y$ with W affine $\Gamma(W, \mathcal{O}_W)$ is a glueable R-module.*

Proof. It is immediate that (2) implies (1). Assume $\{Y_i \to Y\}$ is as in (1) and let $W \to Y$ be as in (2). Then $\{Y_i \times_Y W \to W\}_{i \in I}$ is an étale covering, which we may refine by an étale covering $\{W_j \to W\}_{j=1,\ldots,m}$ with W_j affine (Topologies, Lemma 0218). Thus to finish the proof it suffices to show the following three algebraic statements:

(1) if $R \to A \to B$ are ring maps with $A \to B$ étale and A glueable as an R-module, then B is glueable as an R-module,
(2) finite products of glueable R-modules are glueable,
(3) if $R \to A \to B$ are ring maps with $A \to B$ faithfully étale and B glueable as an R-module, then A is glueable as an R-module.

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Namely, the first of these will imply that $\Gamma(W_j, \mathcal{O}_{W_j})$ is a glueable R-module, the second will imply that $\prod \Gamma(W_j, \mathcal{O}_{W_j})$ is a glueable R-module, and the third will imply that $\Gamma(W, \mathcal{O}_W)$ is a glueable R-module.

Consider an étale R-algebra homomorphism $A \to B$. Set $A' = A \otimes_R R'$ and $B' = B \otimes_R R' = A' \otimes_A B$. Statements (1) and (3) then follow from the following facts: (a) A, resp. B is glueable if and only if the sequence

$$0 \to A \to A_f \oplus A' \to A'_f \to 0, \quad \text{resp.} \quad 0 \to B \to B_f \oplus B' \to B'_f \to 0,$$

is exact, (b) the second sequence is equal to the functor $- \otimes_A B$ applied to the first and (c) (faithful) flatness of $A \to B$. We omit the proof of (2). $\square$